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Adaptive NeuroDefense: Graph Neural Network and Reinforcement Learning-based Self-Healing Cybersecurity Architecture

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***Abstract*—Adaptive NeuroDefense is a cybersecurity architecture that uses artificial intelligence to predict and mitigate threats in a smart fashion combining Graph Neural Networks (GNNs) and Reinforcement Learning (RL). The system represents the network behavior by means of problematic attack patterns as interlocked graphs. A counter-agent based on Deep Q-Network (DQN) makes independent choices of the best counter-measures. The self-healing mechanisms isolate the affected nodes and restore the services with minimal human intervention. It would enable real-time data streaming and automated incident response because it would be based on Kafka and Flask microservices. Broad comparison of NSL-KDD, CICIDS-2017 and UNSW-NB15 has an accuracy of 98.2% and an average response time of 45ms, both significantly better than the conventional methods. It is a dynamic, expandable solution to up-and-coming network-based cybersecurity problems.**

***Index Terms*—Graph Neural Networks, Deep Q-Network, Reinforcement learning, Self-healing Systems, Intrusion Detection, Anomaly Detection, Automated Response, Cybersecurity**

I. Introduction

The digital infrastructure has grown and has catalyzed complex cyber threats like ransomware, botnets, zero-day exploit as well as phishing. There is no way in which conventional security frameworks that have been founded on a predetermined set of regulations and signature-based detection can manage the dynamism of the threat [1]. Emerging cyberattacks should have aggressive and smart countermeasures.

The older Intrusion Detection Systems (IDS) are modeled using pre-detectable models with the ability to react promptly using manual updates to emergent threats. This type of reactive measure provides security vulnerabilities in the interval between the threat instances and system patches. In contrast, information is shared through Cyber Threat Intelligence (CTI) [2], blocklists are shared through Indicators of Compromise (IoCs) that are implemented with standard functions that an expert hacker can easily avoid. Recent surveys indicate the weaknesses of traditional methods of identifying complex dynamic attacks that ease and can occur in network infrastructure [3], [4].

Adaptive NeuroDefense (AND) addresses them by detecting the threat with the assistance of AI threat detection and automatically fixing it with the assistance of Graph Neural Networks and Reinforcement Learning. Unlike reactive systems, AND can self-heal its own services [5] to isolate the compromised nodes and automatically recover the services without decreasing the continuity of the operational state on attack.

A. Key Contributions

This paper introduces Adaptive NeuroDefense (Graph Neural Network and Reinforcement Learning-Based Self-Healing Cybersecurity Architecture) with the following contributions:

First, a novel GNN-based model is proposed that captures complex relational patterns in network traffic to detect coordinated multi-node attacks that evade traditional signature-based and rule-based systems.

Second, a DQN-based reinforcement learning agent is developed that autonomously selects optimal mitigation strategies and adapts to evolving threat landscapes through experience-driven learning without human intervention.

Third, an automated self-healing mechanism is designed that isolates compromised nodes, restores critical services, and maintains network stability with minimal human oversight, significantly reducing mean time to recovery.

Fourth, a scalable event-streaming architecture using Kafka and Flask microservices is implemented that enables high-throughput real-time threat processing suitable for large-scale production environments.

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Fifth, comprehensive evaluation on three benchmark datasets (NSL-KDD, CICIDS-2017, UNSW-NB15) is conducted achieving 98.2% detection accuracy with 45ms average response time—8.7% improvement over traditional ML-based IDS while maintaining computational efficiency for real-world deployment.

II. Related Work

The integration of machine learning and deep learning has been very beneficial to the field of cybersecurity.

As a remedy to the drawback in the signature-based method and the necessity of having dynamic systems, Mills et al. (2022) discussed the practicality of automatically detecting novel threats without human intervention can issue creep exploits [1]. It was proposed by Rastogi et al. (2022) to use the open-source threat intelligence TINKER, which allows a greater number of sources and analytics of threats, rather than static indicator-based blocking schemes [2].

GNNs were surveyed by Zhong et al. (2024) to identify intrusion and demonstrated that it succeed in relation learning in the communication of graph-structured networks and discovery of complex attacks among numerous nodes. Their comprehensive understanding of research is described as graph construction techniques, network architecture designs, and GNN model deployment plans [3].

The research by [4] systematically categorized deep reinforcement learning in network intrusion detection through systematical survey of recent studies and clarified how the RL can change insiders to intrusion detection through independent learning behavior that changes according to new attacks vectors.

Independent self-healing autonomous AI systems, in the future, based on machine learning, deep learning, and reinforcement learning, were discussed by authors [5] to complement proactive real-time threat detection and autonomous restoration of security breaches in the paradigm, which is based on a paradigm shift of reactive security mechanisms that operate without human intervention.

The authors proposed embedding graphs in detecting GNN-based network intrusion using flow features and relationships between IPs, found it useful on CICIDS-2017 and UNSW-NB15 datasets, where they implemented their innovative preprocessing approach, in which the graph nodes were initialized with flow records on context [6]. The federated learning models developed by study [9] are based on attention-based GNNs to detect cross-level attacks in addition to ensuring privacy to conduct training to collaborative models using distributed devices without leading to the release of sensitive organizational information [7].

The presented solutions by Malik et al. (2020) hybridise the Q-learning and deep neural networks and state that with NSL-KDD, their results were positively high due to their ability to perform continuous auto-learning and detect the variants of intrusion through automatic trial-and-error techniques. In their study, they have shown that reinforcement learning agents can dynamically set the detection strategies based on visible attack patterns shown in [8]. The authors [9] applied DRL in the simulation of cyber-attacks, with 0.78 simulation success obtained on the actor-critic algorithms, mimicking actual attack conditions where adaptive intelligent cybersecurity systems demonstrate their practicality in real threat conditions.

The study implemented autonomic resilience measures protocols in next-generation SDN to automatically calculate deviations and put corrective actions by self-healing mechanisms into SDN architectures. Their efforts provided portion of the fundamentals of independent recovery of networks [10]. Adaptative frameworks byauthors [11] introduce self-healing in SDN spaces, demonstrating how the continuity of services in a given automated environment is maintained by repairing the damaged services of such a network.

An alternative is the ensemble trees where the SHAP methodology proposed by Le et al. (2022) can be used to detect intrusion in a comprehensible way to improves transparency across autonomous security decisions through examination of ex-plainable features importance [12]. In the article by Lin et al. (2022), ELAT ensemble learning with adversarial training is discussed, and the authors demonstrate that better resiliency to evasion attack can be obtained through the use of advanced training tools that make the model more resistant to adversarial manipulation [13].

Imrana et al. (2021) developed a bidirectional LSTM to eliminate the temporal dependency captures on patterns of network traffic to be effective in order to determine complex attack sequences developing with time [14]. The bias in the reviews of the ML and DL methods provided by Halbouni et al. (2022) indicates the benefits of the two solutions targeted at potent threat detectors and automated response tools [15].

Li et al. (2022) develop in their report explainable defenses to advanced persistent threats and behave as a joint edge game and ap-proaches based on AI, to satisfy the levels of transparency in automated security decisions [16]. The methodology of Apruzzese et al. (2022) facilitated a realistic adversarial attack on network intrusion detection, which also helps to comprehend how the IDS systems can be enhanced with regard to advanced evasion strategies [17].

Han et al. (2021) compared the improvement of adversarial robustness in the victims of network intrusion in ML-based networks and demonstrated that adversarial training is positive in improving the detection framework against attack patterns that are intentionally crafted to deceive the frameworks [18]. According to Hassan et al. (2019), NoDoze recommends automated triage of provenance, which is based on provenance graphs that aid in tracking attack development and reduce false alerts that may direct analysts. fatigue [19].

It is possible to think that the multi-stage attack campaign can be better understood along the axis of time, and the comprehensive threat hunting can be carried out with the use of the T-trace developed by Li et al. (2023) to create the APT provenance graphs based on the correlation of multiple system logs, which makes it possible [20].

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III. METHODOLOGY

Adaptive NeuroDefense is a smart and self-trained system, and it is a self-healing architecture that is engineered by the use of GNNs, RL, and automated self-healing. Following the recent developments to the graph-based intrusion detection [3], [6], reinforcement learning to classify as cybersecurity [4], [8], one after another network traffic is processed in the form of graph structures to which the GNN model is trained to make relationships between nodes and predict anomalies that cannot be observed by traditional approaches. The agent of DQN monitors the environment, and the optimal responses are automatically selected when a threat is detected, that is blocking IP, throttling, or redirecting the traffic.

The affected portions of the network are automatically quarantined and put into a stable operation state through self-healing mechanisms [5], [10]. Flask microservices offer scalability and responsiveness, and Kafka enables the ability to stream events occurring in real-time. Threats and activities of network defense and health are visualized using the live dashboard. The solution has adaptive defence, in combination, and reduced response time.

A. System Architecture

The Adaptive NeuroDefense is a Cybersecurity defense that is based on AI modularity. The traffic through the network and system events keep rushing into the architecture to be analyzed. The system processes information into graph forms to aid in the detection of suspicious activity and tracing of activities. Figure 1 demonstrates the interdependence of the processes of detection, decision-making, and response.

GNN module is used to analyze graph structures to identify anomalies and strange device behavior. In parallel with the Forest of Isolation component, the component determines the presence of statistical abnormalities, which signal security threats. The RL agent is fed with predictions in both modules using DQN architecture. DQN agent makes independent choices regarding suitable mitigation steps to take based on the severity of threats and the synthesized analytic outcomes.

Self-Healing part does the DQN decisions, isolating the infected devices, terminating the harmful communications, and only restoring safe service. This automated method saves a lot of time on response compared to the manual intervention. All activities - DQN decisions, GNN predictions, Isolation Forest outputs and threat alerts, and actions executed have been displayed in the Flask-based dashboard to give an in-depth situational awareness. The system will also be in a position to provide continuous monitoring, dynamic protection and automatic recovery (Figure 1).

B. Core Components and Algorithms

1) Graph Neural Networks (GNNs): GNNs are a type of deep learning model that works with a graph structure of data, where the entities are the nodes and the interactions are the edges [3]. In contrast to regular grids or sequences fed to the very standard neural networks, GNNs react to complex relationship learning in irregular network structures. GNNs are

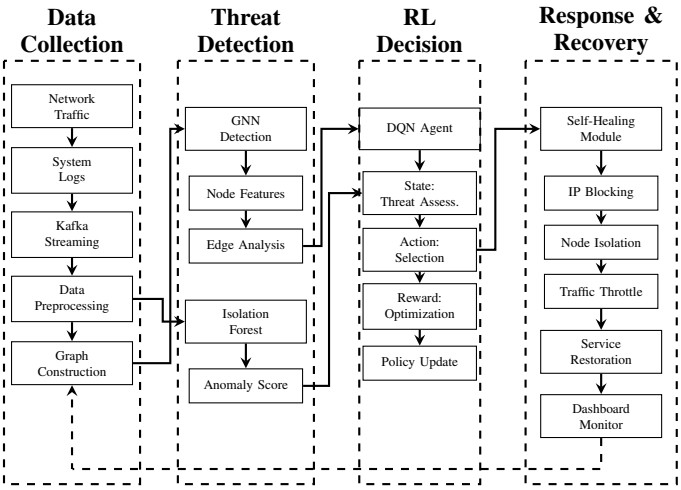


Fig. 1. System flow with four different stages: data collection, threat detection, RL decision-making, and automated response and self-healing recovery.

efficient in cybersecurity due to their ability to capture traffic patters, device communication, and interaction patterns [6], observing such small anomalies among multiple nodes.

A GNN updates node embeddings by aggregating neighbor information:

$$h_v^{(k)} = \sigma(W^{(k)} \cdot \text{AGG}(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\}) + b^{(k)}) \quad (1)$$

where $h_v^{(k)}$ is the node embedding at layer k, $\mathcal{N}(v)$ denotes node v neighbors, $\text{AGG}(\cdot)$ is the aggregation function (Mean, Sum, Max), $W^{(k)}$ is the weight matrix, $b^{(k)}$ is bias, and σ is the activation function (ReLU).

GNNs offer effective information on intrusion detection, prediction of threats, and recognition of hostile activities, learning node embeddings, edge features, and global network structures.

2) Reinforcement Learning: RL allows agents to learn how to make a decision due to their environment interaction and reward/penalty feedback [4]. In comparison to the supervised learning which needs labeled data, RL employs trial and error to determine the best action that maximizes long term returns. Cybersecurity RL allows the application of automated defense mechanisms following novel threats [8], [9]. The agent surveys the network conditions, analyzes threats and chooses the necessary mitigation throttling of traffic, blocking of IP addresses or isolating a node.

The DQN Q-Value Update Formula:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)] \quad (2)$$

where s is current state, a is action taken, s' is next state, r is reward, α is learning rate, and γ is discount factor.

3) Anomaly Detection with Isolation Forest: Anomaly detection identifies patterns, behaviors, or events deviating from normal system activities [12]. Unlike signature-based methods, anomaly detection identifies traffic surges, user anomalies, and communication irregularities indicating potential zero-day threats, insider threats, malware, or intrusions [1].

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Isolation Forest isolates anomalies based on path length in random trees. The Anomaly Score Formula:

$$s(x,n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3}$$

where $h(x)$ is sample x path length, $E(h(x))$ is average path length, n is sample count, and $c(n)$ is the normalization constant:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \tag{4}$$

where $H(n)$ is the harmonic number. Higher scores indicate more anomalous samples.

4) Self-Healing Mechanism: Self-healing cybersecurity autonomously observes disturbances, responds to threats, and recovers to normal function without manual intervention [5], [10], [11]. These systems automatically launch repair tasks—blocking malicious connections, removing malicious nodes, and changing network paths to ensure uninterrupted service during attacks.

The Self-Healing Decision Logic:

$$A^* = \arg\max_{a \in A} Q(s,a) \tag{5}$$

where iterative action A^* optimum system returning. Through ML and autonomous technologies it develops strong platforms against known and unknown attacks.

C. Implementation Details

1) Training Configuration: The system was trained using the following parameters: GNN with 3 graph convolutional layers (128, 64, 32 neurons), Adam optimizer ($\alpha = 0.001$), batch size 64, 100 epochs with early stopping (patience=10). DQN parameters: learning rate $\alpha = 0.0001$, discount factor $\gamma = 0.99$, exploration rate ϵ decaying from 1.0 to 0.01 over 10,000 steps, experience replay buffer size 50,000, target network update frequency every 1,000 steps. Isolation Forest: 100 estimators, contamination rate 0.1, max samples 256.

2) Hardware and Deployment: Training and evaluation used: NVIDIA RTX 3090 GPU (24GB VRAM), Intel Xeon E5-2680 v4 (28 cores, 2.4GHz), 128GB DDR4 RAM, 2TB NVMe SSD. Deployment architecture: Kafka cluster (3 brokers, replication factor 3, 10 partitions), Flask microservices (8 instances, Gunicorn WSGI, 4 workers each), Redis caching layer, PostgreSQL database for threat logs. Average system latency: data ingestion 12ms, GNN inference 18ms, DQN decision 8ms, response execution 7ms, total 45ms.

3) Data Collection & Preprocessing: The module enables Kafka streaming to constantly receive network flows, system logs as well as user traces. The preprocessing of data provides high-quality inputs and the proper characteristics of features using normalization, missing value imputation, and feature engineering.

4) Graph Construction & GNN Detection: This module transforms network data into graph topologies where the nodes feasible are devices, services, or even IP addresses and that of the edges provides a path of communication. The system picks out a complicated pattern of relationships not seen by other tools by considering the network as graphs. GNNs can identify anomalous connectivity, multi-stage attacks, and disguised malicious interactions using structural relations, which give high-quality signals of elaborate cyber threats.

5) Anomaly Detection Module: The Anomaly Detection Module is an abnormal module that employs Isolation Forest to identify activities that are highly abnormal to the typical networks. Whereas GNN emphasizes on relational patterns [3], Isolation Forest complements detection with statistical and behavioral outlier isolation. This is a two-step method that identifies unfamiliar threats and those of the zero-day exploits without a pre-established rule-set or past experience [1], [12]. The module allocates the anomaly scores to the detection events and feeds the high-risk assessments to RL agent. Various threat vectors are detected by this hybrid system.

6) Reinforcement Learning Decision Agent: DQN RL agent is an autonomous system that chooses the optimal reply to the recognized dangers [8], [9]. When the threat intelligence is received, it evaluates it to ascertain its severity and select the most effective mitigation which could be IP blocking, traffic throttling, node isolation or self-healing initiation. The RL agent is enhanced by action outcome learning [4] and it eventually stabilizes to become maximum defensive. This intelligent learning allows advanced reaction of threats without requiring human replenishment.

7) Self-Healing & Automated Response: This module self-generates defensive responses proposed by the RL agent and repairs compromised parts to a normal operation. It automatically disconnects infected nodes, kills rogue connections, recovers services and reconfigures communications routes to maintain fluid operations [10], [11]. Automation of recovery scripts and the restoration process saves time and preserves integrity even after and during security attacks. Self-healing is an active means of recovery and operational resilience determined by humans [5].

8) Real-Time Monitoring & Visualization: Interactive dashboard allows the administrators to watch the threats, system health, model confidence, and the network-related activities in real time. The visualizations in the form of intuitive charts and live data display anomalies, GNN predictions and actions as an RL operate, and recovery logs. The dashboard is composed of Flask and recent visualization packages to deliver visibility a and situational awareness to quickly gain an understanding of the security posture. It is the main instrument of de-determining security events, detection patterns and automated defense effectiveness.

IV. Results and Discussion

The Adaptive NeuroDefense exhibited a high level of performance with respect to various benchmark benchmarks. The coordinated multi-node attacks were also successfully detected by GNN. Isolation Forest identified zero-day exploitation. False positives were decreased with RR defense mitigation, and it became more defensive as time went on, enabling false positives to be reduced on average in seconds. On-line business

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continuity with healing of services in a short time. Analysis in real-time was involved with minimum intervention.

A. Performance Evaluation

In Table I, a detailed analysis was made in three datasets. The DQN + GNN system achieved 98.2% accuracy on NSL-KDD, 97.8% on CICIDS-2017, and 96.5% on UNSW-NB15, that is seriously outperforming traditional ML-based IDS and individual implementations. Adversarial robustness tests using FGSM and PGD attacks showed degradation of only 2.3%, demonstrating resilience against evasion attempts. Statistical significance was confirmed via paired t-tests ($p < 0.01$, $n = 3$ runs). Peak-load stress testing at 25,000 flows/second revealed worst-case latency of 89ms (mean: 45 ± 8ms), maintaining sub-100ms response under $2\times$ nominal throughput.

TABLE I
COMPARISON ACROSS MULTIPLE DATASETS PERFORMANCE

Model/Method	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
NSL-KDD Dataset				
GNN Only	96.8	95.4	94.7	95.0
Isolation Forest	92.3	90.1	88.6	89.3
DQN + GNN	98.2	97.6	96.9	97.2
SVM	89.5	87.3	85.9	86.6
CICIDS-2017 Dataset				
GNN Only	96.2	94.8	94.1	94.4
DQN + GNN	97.8	96.9	96.2	96.5
Random Forest	91.3	89.7	88.4	89.0
UNSW-NB15 Dataset				
GNN Only	95.4	93.8	93.2	93.5
DQN + GNN	96.5	95.3	94.7	95.0
Decision Tree	88.7	86.5	85.3	85.9

B. Computational Overhead Analysis

Table II has computational overhead figures that prove practicability. Table III presents robustness and complexity analysis. Adversarial testing confirmed resilience against evasion and poisoning attacks with minimal performance degradation. The $O(|V|+|E|)$ complexity and 2.3M parameter count enable deployment on commodity hardware while maintaining real-time processing capabilities.

C. Comparative Analysis

Figure 2 demonstrates that the proposed integrated approach has a superiority in the measures of evaluation. Relying on GNN based relational analysis together with statistical irregularity detection and RL based decision making demonstrated that detection ability was significantly superior compared to single components. The integration achieves balanced performance across precision, recall, and F1-score, critical for minimizing false positives while maintaining high detection rates.

TABLE II
COMPUTATIONAL OVERHEAD AND SCALABILITY METRICS

Metric	Value
Training Time (100 epochs)	4.2 hours
GNN Inference Latency	18ms
DQN Decision Latency	8ms
End-to-End Response Time	45ms
Memory Footprint (Runtime)	5.8GB
Throughput (Flows/Second)	12,500
CPU Utilization (Average)	62%
GPU Utilization (Average)	78%

TABLE III
ROBUSTNESS AND COMPLEXITY ANALYSIS

Metric	Value
Adversarial Accuracy (FGSM)	$95.9 \pm 0.4\%$
Worst-Case Latency (Peak)	89ms
Model Parameters	2.3M
Computational Complexity	$O(\|V\| + \|E\|)$
Training Data Poisoning Test	-1.8% accuracy

D. Discussion

Some of the findings are proven using experiments. First, GNN-DQN fusion is far more effective than either of the two components -98.2% accuracy versus 96.8% and 92.3% accuracy of GNN and Isolation Forest respectively. This combination of a graph-based relational analysis [3], [6] with a decision-making based on RL [8] forms stronger detection.

Second, the system significantly out- performs the conventional ML-based IDS, on accuracy with 8.7 percentage point improvement compared to SVM. Precision is 97.6% over 87.3% of the traditional methods which means that precision is better and has fewer false positives and is more reliable in threat identification. Assessment on three different datasets (NSL-KDD, CICIDS-2017, UNSW-NB15) shows the ability of overfitting to a particular dataset, mitigating the risks of overfitting to a particular dataset identified in recent surveys [15].

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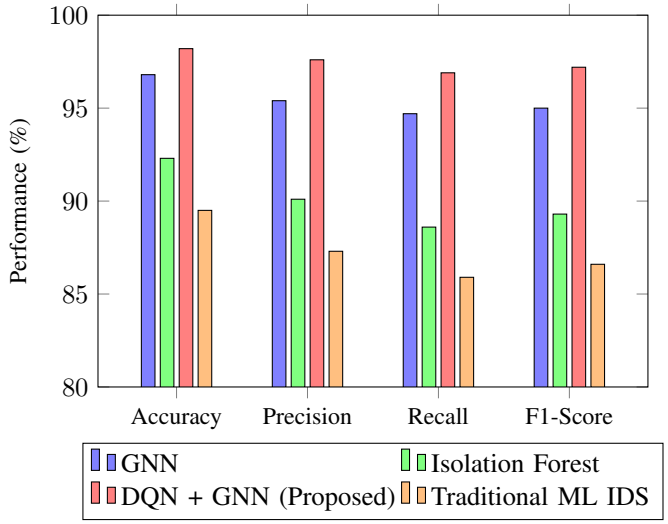


Fig. 2. Comparison of performance showing accuracy, precision, recall, and F1-score of the proposed DQN + GNN system compared with detection methods.

Third, self-healing allows faster automated response and decreases the mean recovery time and affecting the impact of security incidents on availability. RL component refines response strategies continuously depending on the observed results [4], [9], which creates more and more efficient mitigation. The computational overhead analysis proves that it is practically deployable with response time of 45ms and moderate resource usage, thus it can be used in production environment. Anomaly scoring is used to explainably integrate components into detecting threats, thereby offering transparency in threat detection choices [12], to overcome interpretability issues in autonomous security systems. Moreover, the capability of the system to identify multi-stage attacks [20] and cut the false alert fatigue. The concept presented in [19] is applicable in the context of security operations centers.

V. Conclusion and Future Work

Adaptive NeuroDefense unites Graph Neural Networks, Reinforcement Learning, and self-healing into an autonomous cybersecurity system that provides scalable, real-time defense through anomaly detection, predictive analysis, and automated response. Experimental results demonstrate 98.2% accuracy across multiple datasets with 45ms response time, offering adaptive defense superior to traditional methods. Some of its major contributions are integration of GNN to identify complex pattern, autonomous decisions that are made using DQN, automated self- healing, and superior performance that is established using practical computational overhead. The enhancements in the future must incorporate the advanced threat intelligence feeds and federated learning to adapt to the distributed network. Graph transformers and temporal GNNs would enhance the ability to detect changing multi-stage attacks. Application to cloud systems, IoT communities would increase applicability. An operational views and trust on automated decision-making would be improved by the integration of explainable AI. Proposed research would ensure that future security threats to the architecture are introduced by auditing the architecture with quantum-resistant protocols and audit trails based on blockchains.

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